Measurement IS Thermodynamics

A Machine-Checked Proof Connecting Wave-Particle Duality
to Landauer's Erasure Bound

UMST Formal Methods

March 2026

Core Result

Principle of Maximal Information Collapse. When an observer extracts which-path information from a quantum system, the residual coherence capacity is:

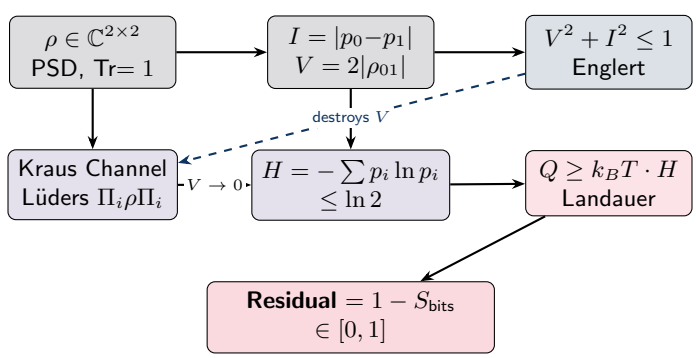
$$\boxed{\text{Residual} = 1 - \frac{\text{MI}_{\text{extracted}}}{k_B T \ln 2} \in [0,\, 1]}$$

Extract 0 bits $\implies$ full interference. Extract 1 bit $\implies$ complete decoherence.

Machine-checked in Lean 4 with Mathlib. ***360 theorems, 0 sorry, 1 physical axiom.***

1 The Proof Architecture

We formalize the double-slit experiment as a *closed chain* from quantum states through measurement to thermodynamic cost, proving that observation and decoherence are mathematically identical.



2 What Is Formally Proved

3 The Physical Argument

A particle traverses a double-slit apparatus. Its spatial path qubit is a 2×2 density matrix ρ.

1. **No detector.** Off-diagonals $\rho_{01} \neq 0$ carry coherence; interference fringes appear ($V > 0$).
2. **Which-path detector.** The Lüders channel $\mathcal{E}(\rho) = \Pi_0 \rho \Pi_0 + \Pi_1 \rho \Pi_1$ zeros off-diagonals: $V \to 0$. Englert guarantees I rises accordingly.

Theorem	Statement	Module
Englert complementarity	$V^2 + I^2 \leq 1$	`QuantumClassicalBridge`
Which-path collapse	$V \to 0$ after Lüders channel	`MeasurementChannel`
Projector properties	self-adjoint, idempotent, orthogonal, TP	`MeasurementChannel`
Diagonal entropy bound	$H_{\text{diag}} \leq \ln 2$	`InfoEntropy`
Landauer cost cap	cost $\leq k_B T \ln 2$	`LandauerBound`
Path entropy ≤ 1 bit	$S_{\text{bits}} \in [0, 1]$	`LandauerBound`
Maximal collapse	$S_{\text{bits}} = 1 \implies \text{Residual} = 0$	`LandauerBound`
Null preservation	$S_{\text{bits}} = 0 \implies \text{Residual} = 1$	`LandauerBound`
Cost = energy × extracted	$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$	`LandauerBound`
Erasure ≥ bound	`dissipatedHeat` ≥ `landauerCostDiagonal`	`LandauerBound`
Which-path invariance	Landauer cost unchanged by measurement	`LandauerBound`
Density matrix diagonals	PSD $\implies p_i \geq 0, \sum p_i = 1, p_i \leq 1$	`DensityState`
Gate enforcement	admissibility + Landauer + cap in one	`DoubleSlit`

3. **Thermodynamic cost.** The observer's memory now holds a classical bit. Erasing it against a bath at T dissipates $\geq k_B T \ln 2$ joules (Landauer). This energy is *exactly* the cost of destroying coherence.

4. **Maximal collapse.** Normalizing: if the observer extracts the full 1-bit capacity, residual coherence $= 0$. Partial extraction leaves partial visibility. The bound is tight at both endpoints.

Conclusion. The observer does not "collapse reality" by an act of consciousness. The observer pays an irreversible thermodynamic cost that is *mathematically identical* to decoherence. **Measurement IS thermodynamics.**

4 Verification Stack

Layer	Tool	Scope	Status
Formal proof	Lean 4 + Mathlib	38 modules, 360 theorems	**0 sorry**
Property testing	Haskell QuickCheck	14 properties + sanity	**All pass**
Numerical sim	Python (stdlib + QuTiP)	54 unit tests, 4 sims	**All pass**
Cross-language	Coq + Agda	LandauerEinsteinBridge	**0 Admitted**

Repository: `umst-formal-double-slit` — *License:* MIT